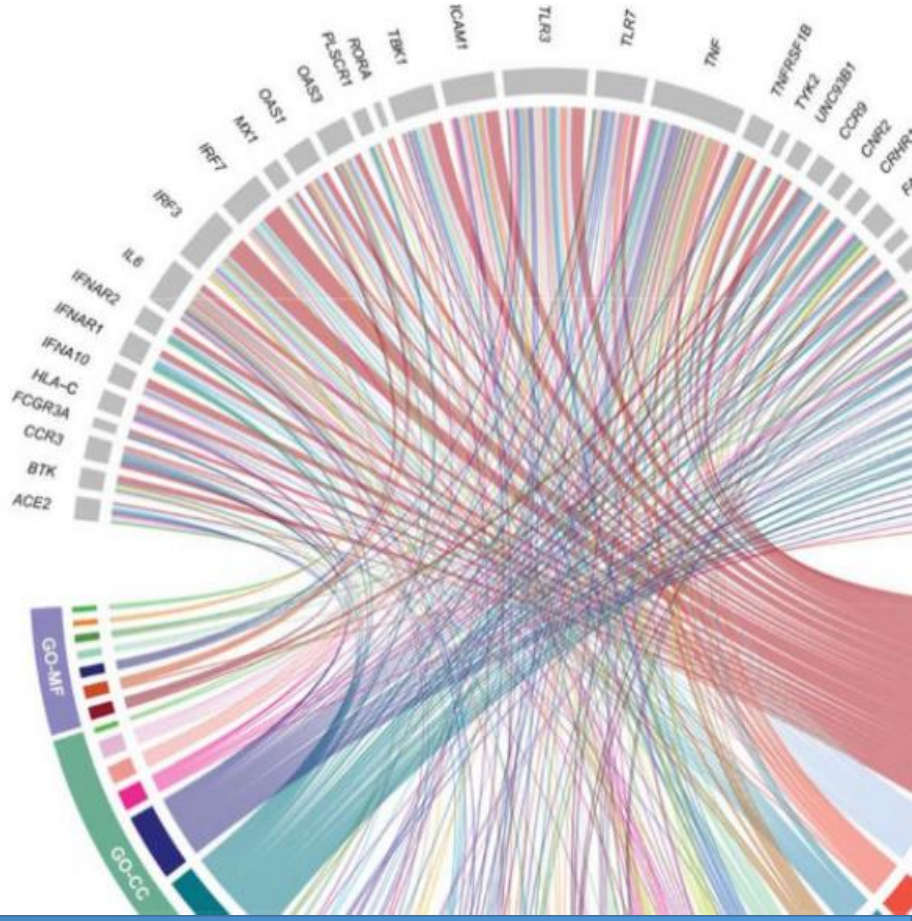
Tècniques i Eines Bioinformàtiques

Exact Matching Problem



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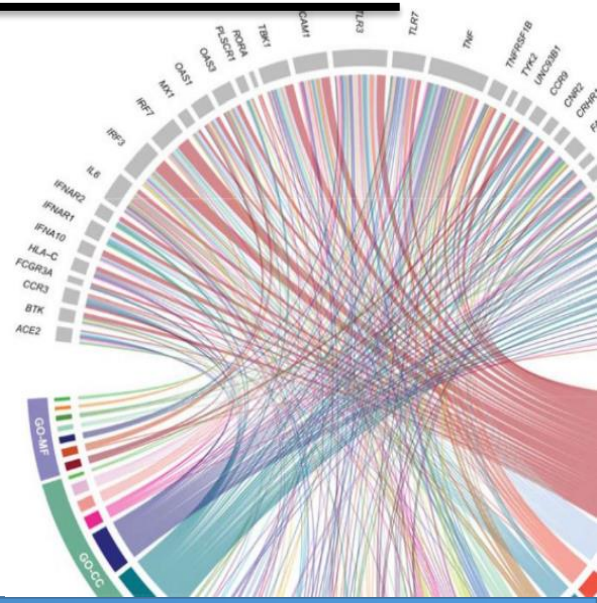
Acknowledgements

Many pictures and materials are taken from **Ben Langmead's course**.

Course heavily inspired in:

- **Genome-Scale Algorithm Design**. Veli Mäkinen, Djamal Belazzougui, Fabio Cunial, Alexandru I. Tomescu. Cambridge University Press.
- **Algorithms on Strings, Trees, and Sequences**. Dan Gusfield. Cambridge University Press.
- **An Introduction to Bioinformatics Algorithms.** Neil C. Jones, Pavel A. Pevzner. MIT Press.

1 Strings

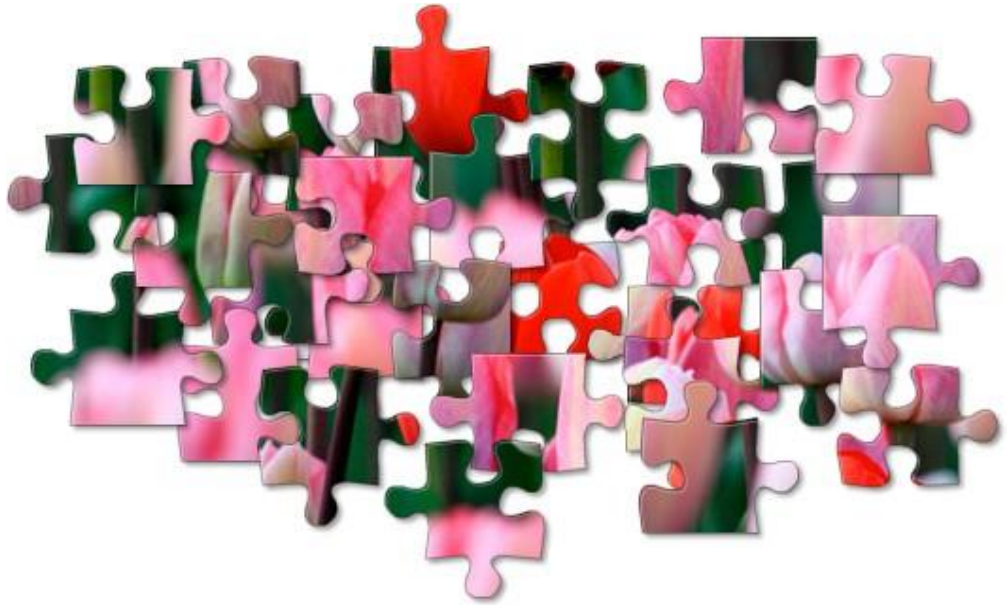

ACE2
BTK
CCR3
FCGR3A
HLA-C
IFNA10
IFNAR1
IFNAR2
IL6
IRF3
IRF7
MX1
OAS1
OAS3
FLSCR1
RORA
TBK1
ICAM1
E2F1
TLR7
TNF
TNFRSF1B
TYK2
UNC93B1
CCR9
CNR2
GO-MF
GO-CC

Sequencing Reads are Strings

```
GTATGCACGCGATAG
TAGCATTGCGAGACG
TGTCTTTGATTCCTG
GACGCTGGAGCCGGA
TATCGCACCTACGTT
CACGGGAGCTCTCCA
GTATGCACGCGATAG
GCGAGACGCTGGAGC
CCTACGTTCAATATT
GACGCTGGAGCCGGA
TATCGCACCTACGTT
CACGGGAGCTCTCCA
```

```
TATGTCGCAGTATCT
GGTATGCACGCGATA
CGCGATAGCATTGCG
GCACCCTATGTCGCA
CAATATTCGATCATG
TGCATTTGGTATTTT
ACCTACGTTCAATAT
CTATCACCCTATTAA
GCACCTACGTTCAAT
GCACCCTATGTCGCA
CAATATTCGATCATG
TGCATTTGGTATTTT
```

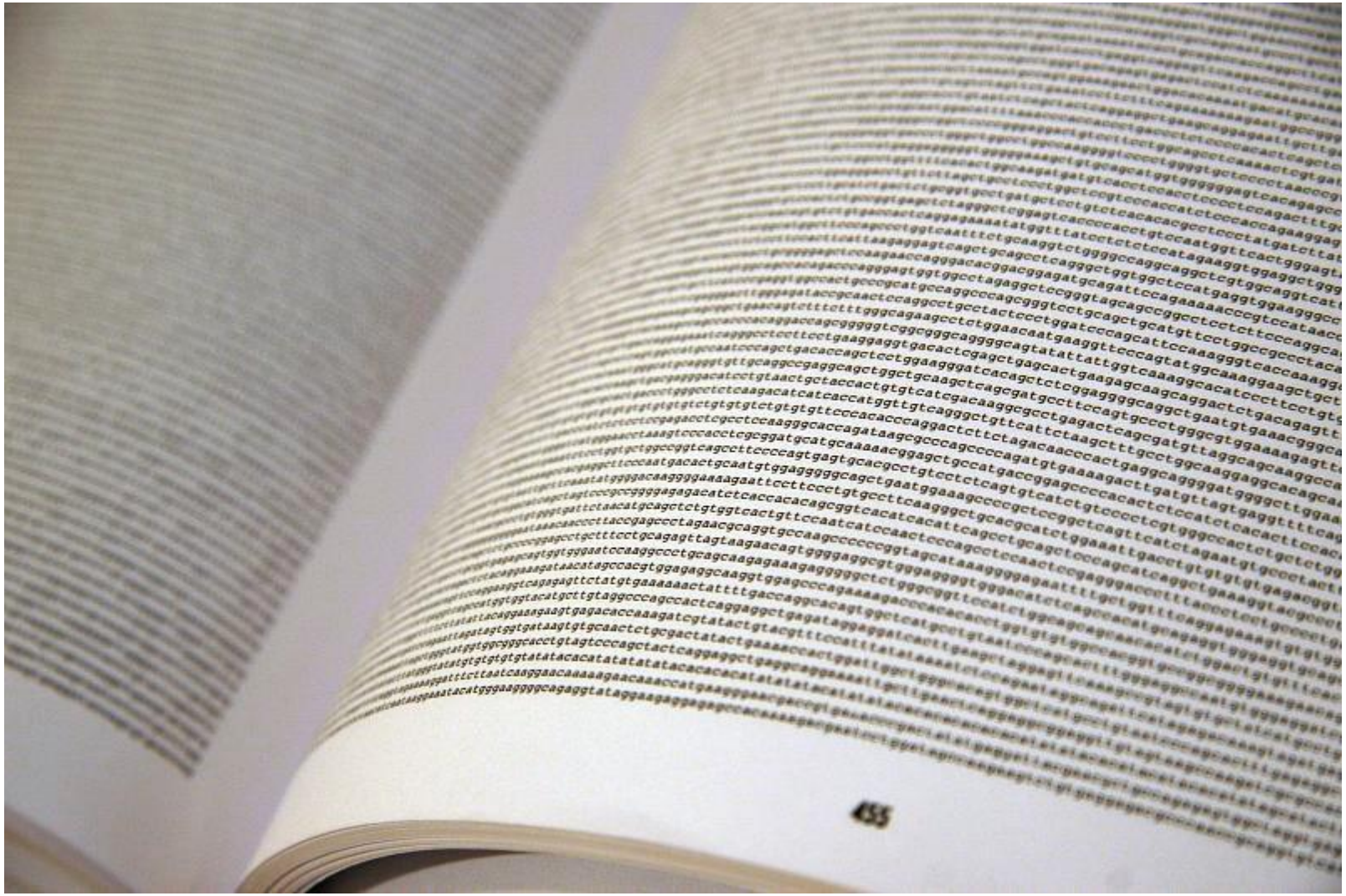
```
CACCCTATGTCGCAG
TGGAGCCGGAGCACC
GCATTGCGAGACGCT
GTATCTGTCTTTGAT
GATCACAGGTCTATC
CGTCTGGGGGGTATG
TATTTATCGCACCTA
CTGTCTTTGATTCCT
GTCTGGGGGGTATGC
GTATCTGTCTTTGAT
GATCACAGGTCTATC
CGTCTGGGGGGTATG
```



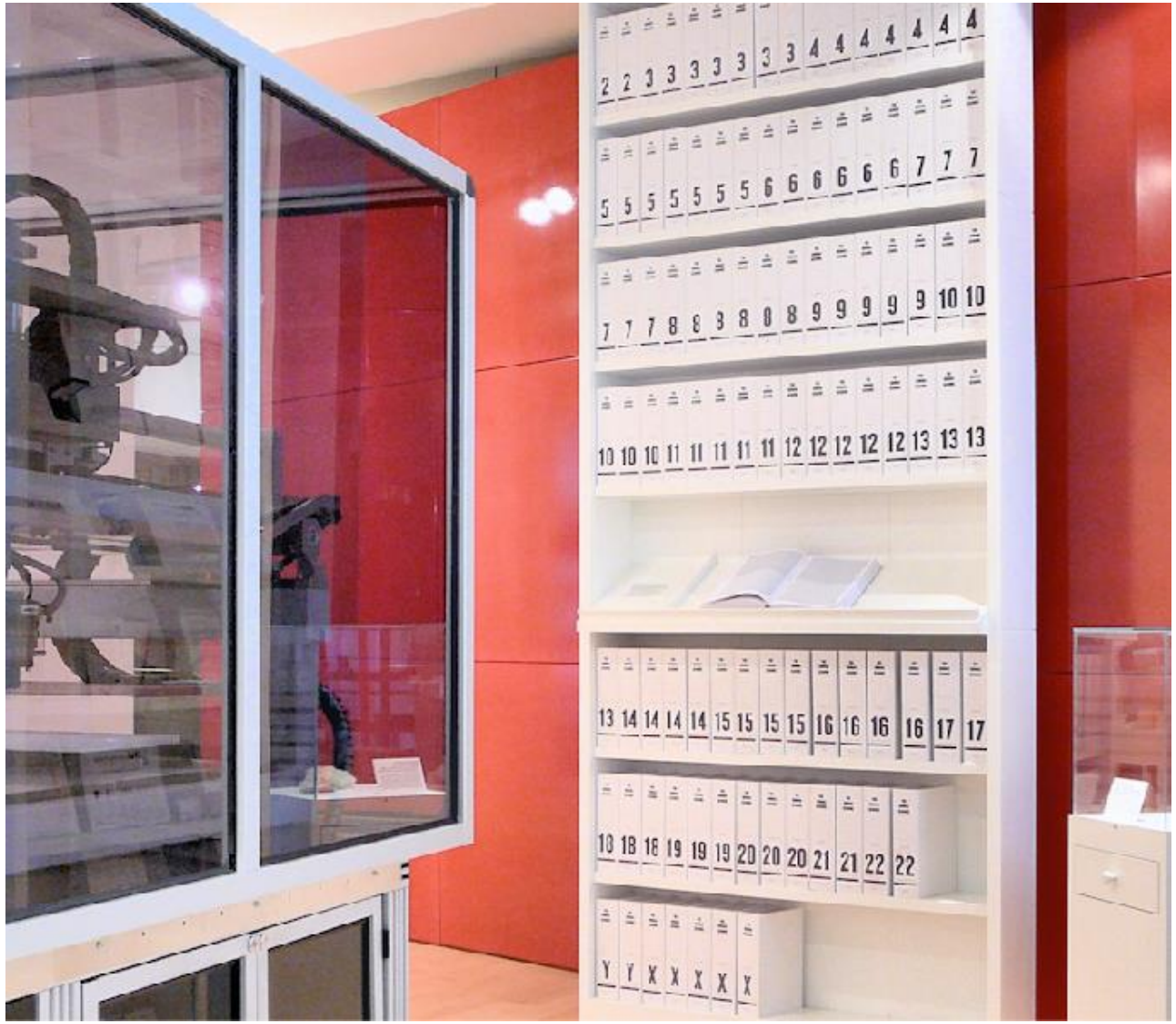
Sequencing reads are *strings;* sequences of characters

The strings are the only hints we get about *where the reads came from from* with respect to the longer DNA molecules...

... like pictures on puzzle pieces



What if I told you to find all the places where the string GATACCA occurs in here?



What if I told you to find all the places where the string GATACCA occurs in here?

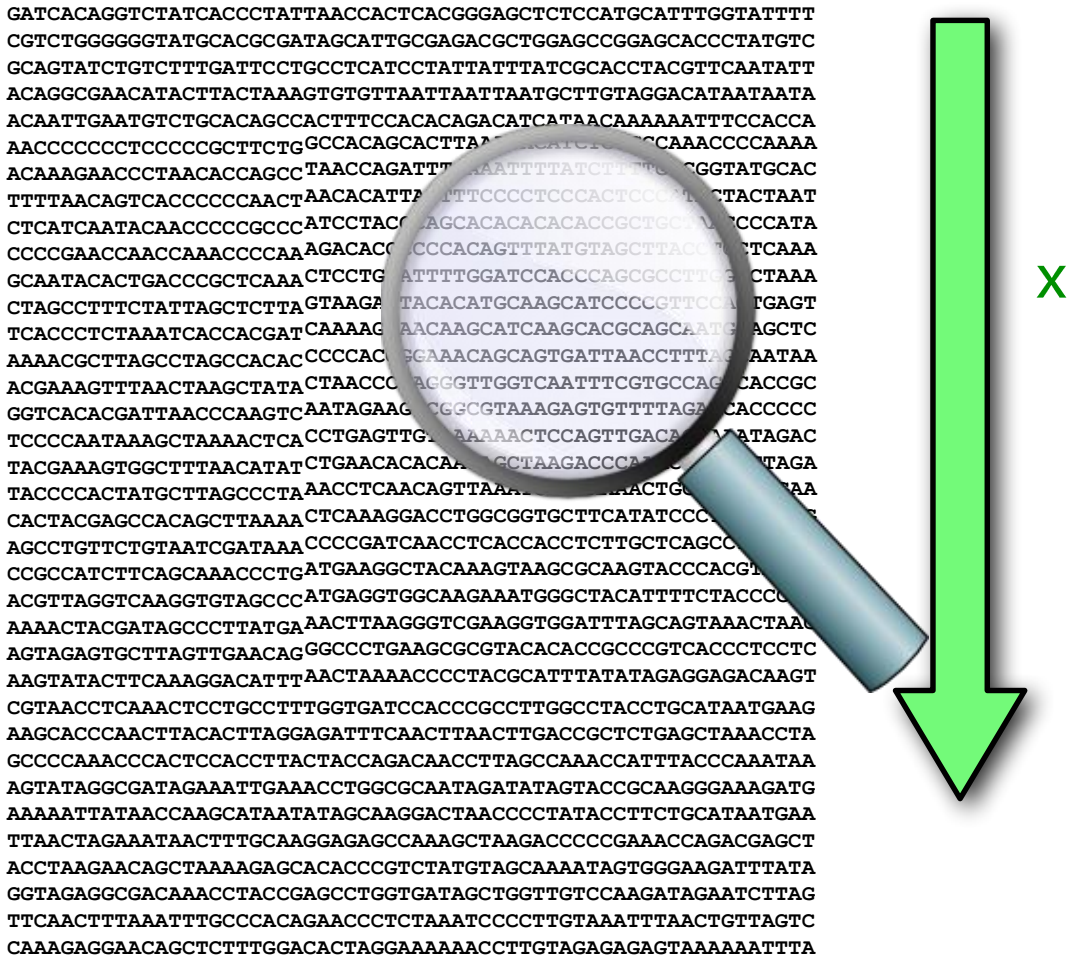
Strings

Read

CTCAAACTCCTGACCTTTGGTGATCCACCCGCCTAGGCCTTC

x billions

Reference



x million

We're going to *need* the right algorithms...

Strings Algorithms and Data Structures are Well Studied

Many kinds of data are string-like: books, web pages, files on your hard drive, medical records, chess games, ...

Algorithms for one kind of string are often applicable to others:

- Regular expression matching can find files on your filesystem (grep), or bad network packets (snort)
- Indexes for books and web pages (inverted indexing) can be used to index DNA sequences
- Methods for understanding speech (HMMs) can be used to understand handwriting or identify genes in genomes

Strings Come from Somewhere

Processes that give rise to real-world strings are complicated. It helps to understand them.

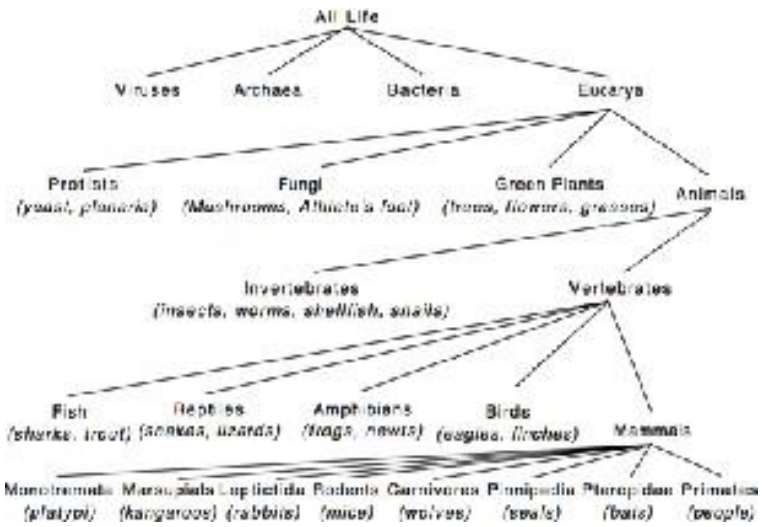
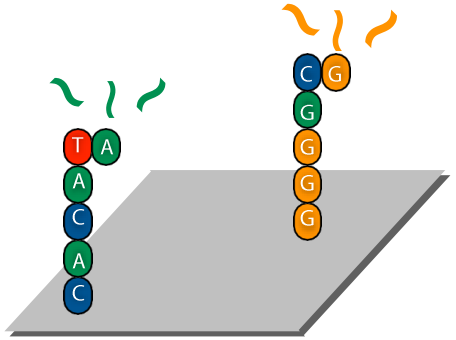


Figure from: Hunter, Lawrence. "Molecular biology for computer scientists." *Artificial intelligence and molecular biology* (1993): 1-46.

1. Evolution:
   - Mutation
   - Recombination
   - (Retro)transposition

2. Lab procedures:
   - PCR
   - Cell line passages



3. Sequencing:
   - Fragmentation bias
   - Miscalled bases

Strings have Structure

One way to model a string-generating process is with coin flips:

{ = A, = C, = G, = T }

But such strings lack internal patterns ("structure") exhibited by real strings

> 40% of the human genome is covered by *transposable elements*, which copy-and-paste themselves across the genome and mutate

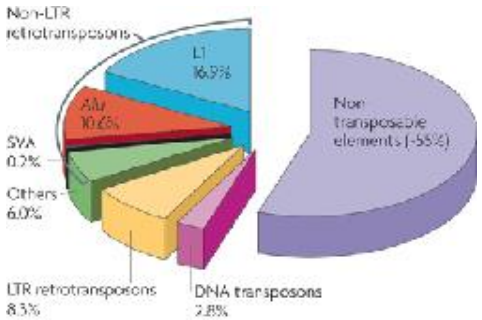


Image from: Cordaux R, Batzer MA. The impact of retrotransposons on human genome evolution. Nat Rev Genet. 2009 Oct;10(10):691-703

Slipped strand mis-pairing during DNA replication results in expansion or retraction of simple (*tandem*) repeats

••• ATATATATATATAT •••

↕

••• ATATATATATATATAT •••

String Definitions

String S is a finite sequence of characters

Characters are drawn from alphabet Σ Usually, Σ = { A, C, G, T }

| S | = number of characters in *S*

```
>>> s = 'ACGT'
>>> len(s)
4
```

ε is "empty string" | ε | = 0

```
>>> len('')
0
```

Positions within a string *S* are referred to with *offsets*

```
>>> s = 'ACGT'
>>> s[0]
'A'
>>> s[2]
'G'
```

Usually assume alphabet Σ is finite, with O(1) elements

Nucleic acid alphabet: { A, C, G, T }

Amino acid alphabet: { A, R, N, D, C, E, Q, G, H, I, L, K, M, F, P, S, T, W, Y, V }

Occasionally we'll consider what happens as |Σ| grows

String Definitions

Concatenation of S and T , ST = characters of S followed by characters of T

```
>>> s = 'AACC'
>>> t = 'GGTT'
>>> s + t
'AACCGGTT'
```

Substring of S is a string occurring inside S

```
>>> s = 'AACCGGTT'
>>> s[2:6]
'CCGG' # substring of seq
```

S is a *substring* of T if there exist (possibly empty) strings u and v such that $T = uSv$

String Definitions

Prefix of *S* is a substring starting at the beginning of *S*

```
>>> s = 'AACCGGTT'
>>> s[0:6]
'AACCGG' # prefix
>>> s[:6] # same as above
'AACCGG'
```

S is a *prefix* of *T* if there exists a string *u* such that $T = Su$

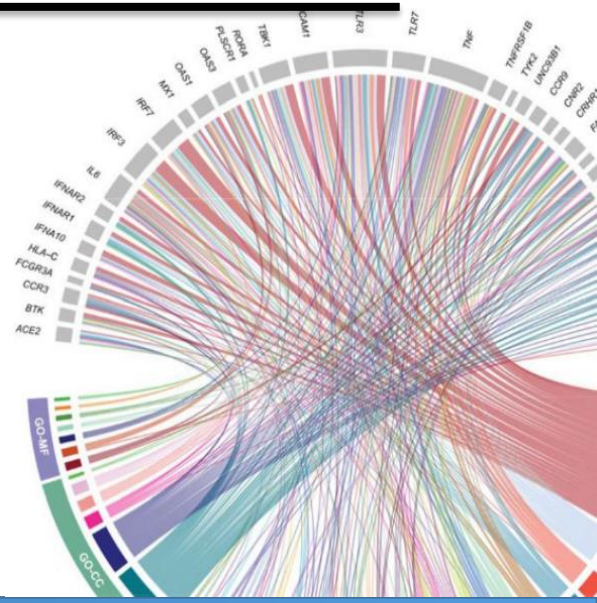
String Definitions

Suffix is substring ending at end of *S*

```
>>> s = 'AACCGGTT'
>>> s[4:8]
'GGTT' # suffix
>>> s[4:] # like s[4:len(s)]
'GGTT'
>>> s[-4:] # like s[len(s)-4:len(s)]
'GGTT'
```

S is a *suffix* of *T* if there exists a string *u* such that $T = uS$

2 Exact Matching



Exact Matching Problem

Find places where *pattern P* occurs as a substring of *text T*.
Each such place is an *occurrence* or *match*.

Let $n = |P|$, and let $m = |T|$ Assume $n \leq m$

Alignment: a way of putting *P's* characters opposite *T's*.
May or may not correspond to an match.

```
P:  word
T:  There would have been a time for such a word
                 word                        word
```

Alignment 1: word Alignment 2: word

Exact Matching Problem

What's a simple algorithm for exact matching?

P: **word**

T: **There would have been a time for such a word**
word word word word word word word word **word**
word word word word word word word word
word word word word word word word word
word word word word word word word word
word word word word word word word word

One occurrence

Try all possible alignments. For each, check if it matches.
This is the *naïve algorithm*.

Naïve Exact Matching Algorithm

Naïve Exact Matching Algorithm

```
def naive(p,t):
  occurrences = []
  for i in range(len(t)-len(p)+1): # Loop over positions (of T)
    match = True
    for j in range(len(p)): # Loop over characters (of P)
      if t[i+j] != p[j]:
        match = False # Mismatch found! (reject alignment!)
        break
    if match:
      occurrences.append(i) # All chars matched (accept alignment!)
  return occurrences
```

Even more naïve: remove **break**

P: **word**

T: **There would have been a time for such a word**

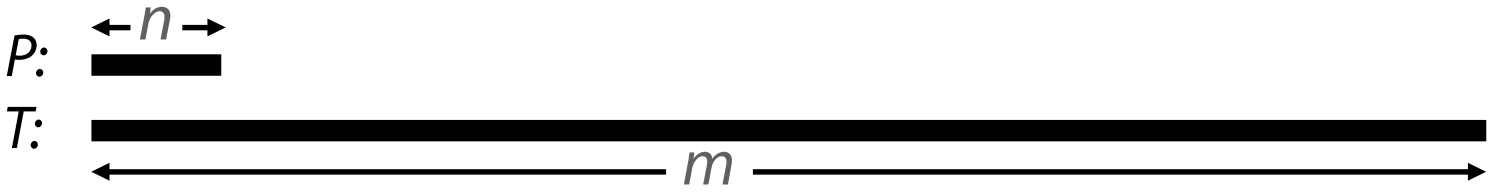
word **word** **word**

Naïve Exact Matching Algorithm

$n = |P| \qquad m = |T|$

How many alignments are possible?

$$m - n + 1$$



Naïve Exact Matching Algorithm

$n = |P| \qquad m = |T|$

Greatest # character comparisons possible?

$$n(m - n + 1)$$

P: aaaa

T: aaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaa

aaaa aaaa aaaa aaaa aaaa aaaa aaaa aaaa aaaa

aaaa aaaa aaaa aaaa aaaa aaaa aaaa aaaa

aaaa aaaa aaaa aaaa aaaa aaaa aaaa aaaa

aaaa aaaa aaaa aaaa aaaa aaaa aaaa aaaa

aaaa aaaa aaaa aaaa aaaa aaaa aaaa aaaa

Naïve Exact Matching Algorithm

$n = |P| \qquad m = |T|$

Least # character comparisons possible?

$$m - n + 1$$

P: **abbb**

T: **bbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbb**

abbb abbb abbb abbb abbb abbb abbb abbb abbb

abbb abbb abbb abbb abbb abbb abbb abbb

abbb abbb abbb abbb abbb abbb abbb abbb

abbb abbb abbb abbb abbb abbb abbb abbb

abbb abbb abbb abbb abbb abbb abbb abbb

Naïve Exact Matching Algorithm

How many character comparisons in this example?

```
P: word
T: There would have been a time for such a word
   word word word word wordword wordword word
    word word word word word word word word
     word word word word word word word word
      word word word word word word word word
       word word word word word word word word
```

Hint: there are 41 possible alignments

Naïve Exact Matching Algorithm

How many character comparisons in this example?

P: `word`

T: **There would have been a time for such a word**

word word word word word word word word word

word word word word word word word word

word word word word word word word word

word word word word word word word word

word word word word word word word word

40 mismatches + 6 matches = 46 character comparisons

Closer to the minimum (41) than the maximum (164)

Exact Matching. Are there Better Algorithms?

P: `word`

T: `There would have been a time for such a word`

`word`

u doesn't occur in *P*, so skip next two alignments

P: `word`

T: `There would have been a time for such a word`

`word`

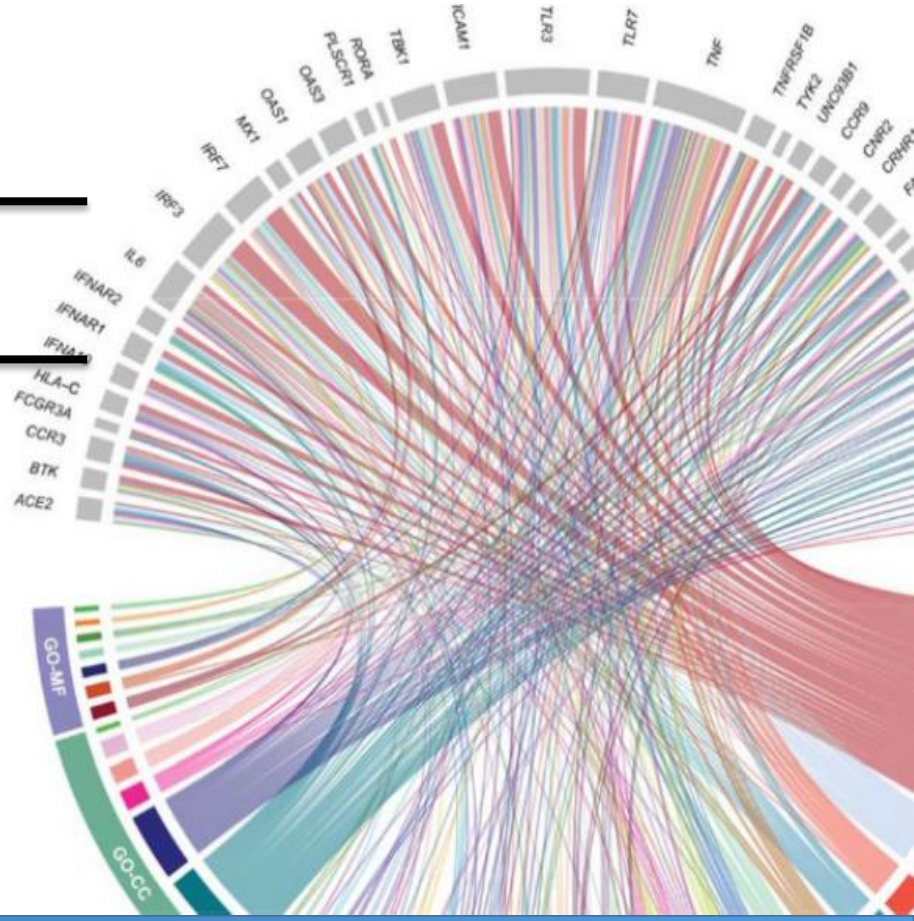
`word` skip!

`word` skip!

`word`

We'll take such ideas further when we discuss Boyer-Moore

Questions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IFNAR2
IL6
IRF3
IRF7
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